

[Problems](#)[Status](#)[Standings](#)

Coins

🏷 Tags:

Time Limit: 1 s Memory Limit: 64 MB
Submission: 30 AC: 8 Score: 99.62

[Submit](#)[Codes](#)[AI Code Tips](#)

Description

It's a problem about coins.

You have n ($1 \leq n \leq 10$) types of coins. Each coin has different value, a_i . The number of each coin you has is b_i . Then you want to buy something which price is p ($1 \leq p \leq 10000$). You are so lazy that you want to go out with coins as less as possible. And you think carrying coins to home is tired. So you only carry coins the sum of which's value equal to p . If your all coins' summary less than p , output "I am so poor!", else if you can't carry coins which value equals to p , output "How bad!", else output the minimum coins you can carry.

Input

There are many testcases.

In each testcase, the first line is n and p , the next line has n integers, a_i ($1 \leq a_i$), the next line has n integers b_i ($1 \leq b_i \leq 100$)

Output

Each testcase output one line about your answer.

Samples

input

[Copy](#)

```
1 10
2
6
1 20
2
1
2 20
5 3
3 2
```

output

[Copy](#)

```
5
I am so poor!
How bad!
```

Author

YU, Dongwei

Submit

Codes

HZNUOJ is based on HUSTOJ

--- Maintainer List ---

Server Time: 2025-8-26 15:39:43